

Bryan Cueva

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA

Master of Science in Mechanical Engineering, Robotic and Control Systems Focus | GPA: 3.53

December 2025

Bachelor of Science in Mechanical Engineering

May 2024

Coursework: Modern Control Theory • Electromechanical Systems • Applied FEA • Space Robotics • Soft Robots
CV-Based Biomechanics • Advanced Statistics • Machine Learning • AI/ML Deployment

Honors: GEM Employer Fellowship (Fully funded program with AFRL sponsorship)

EXPERIENCE

R&D Product Development Engineering Co-Op | Advanced Optonics (Start-up) | Pittsburgh, PA Jan 2026 – Current

- Designed DFM/DFA-driven tooling and fixtures with SolidWorks and SLA prototyping to scale bio-MEMS surgical implant sensors, increasing assembly yield from 30% to 90%
- Applied tolerance stack-up and GD&T principles to reduce alignment failure modes and improve repeatability
- Automated chip-to-PCB alignment using OpenCV object tracking, reducing manual assembly time by 20%
- Built electrochemical test platform for automated DOE, process characterization, and sensor release testing
- Integrated Python DAQ, camera, LCR, and potentiostat into GUI, improving test repeatability and data integrity

R&D Engineering Intern | Air Force Research Lab | Dayton, OH

Jun 2024 – Nov 2024

- Designed custom soft robotic end-effectors and test hardware for contact-rich grasping of non-uniform surfaces
- Performed nonlinear structural FEA in Ansys to model gripper deformation and guide design iterations
- Developed LabVIEW control and DAQ workflow for 8-DOF actuation testing, reducing pressure error by 20%
- Executed single- and multi-DOF gripper performance tests across irregular shapes, improving grasp success by 60%

Manufacturing Engineering Intern | Medtronic, Inc. | Minneapolis, MN

Jun 2023 – Aug 2023

- Led IQ/OQ/PQ and DMAIC root cause analysis for automated equipment, increasing line capacity by 40%
- Optimized machine parameters using DOE in Minitab, reducing defects by 25% and saving \$20k annually
- Created SOPs and tooling updates to improve process reliability in regulated medical device manufacturing

PROJECTS

Haptics Research Assistant | SHRED Lab, Robotics Institute | Pittsburgh, PA

Jun 2025 – Dec 2025

- Designed electromechanical test rig with magnetic actuators and load cells for human-robot haptic characterization
- Implemented Arduino/C++ signal filtering and calibration workflow to improve signal quality and repeatability
- Conducted 30-participant study and analyzed slip detection data across surface conditions using Python/Pandas
- Contributed to accepted IEEE Haptics Symposium WIP paper and poster on tactile slip perception

Lizard-Inspired Robotic Leg | Bioinspired Robotics

Jan 2025 – May 2025

- Led 30+ part test rig design in SolidWorks, integrating IMU, load cell, FDM and machined components
- Optimized four-bar linkage trajectory in MATLAB to model basilisk lizard's aquatic locomotion dynamics
- Analyzed force and motion data in Python using ANOVA to compare test conditions and validate simulations

Flexible Three-Axis Strain Sensor | Soft Robots

Aug 2024 – Dec 2024

- Developed elastic, polymer strain sensors by optimizing material composition, increasing sensitivity by 40%
- Characterized coupled mechanical-electrical response through tensile, compression, and 100-cycle fatigue testing

Portable Wood Bending Machine | Mechatronic Systems

Aug 2023 – Dec 2023

- Built electromechanical device to bend wood with linear actuation, load cell feedback, and Arduino PID control
- Validated Solidworks FEA with hand calcs to determine load requirements and size actuators on a \$450 budget
- Achieved 130 mm deflection in 40 seconds while maintaining position error under 10%

LEADERSHIP

Event Coordinator | Latino Student Association at CMU

May 2022 – May 2025

- Led planning of 30+ events for 300+ members, managed a \$9K budget, and increased weekly attendance by 35%
- Organized CMU's largest Latino cultural event with local vendors and over 500 attendees

SKILLS

Software: SolidWorks (CAD, FEA), Ansys Mechanical, Fusion 360, Python (Pandas), MATLAB, Simulink, C++

Hardware: Arduino, 3D Printing (FDM, SLA), Soldering, Laser Cutting, Machining, Instron, LCR